

Ilia Chelak

Ph.D. student at UTokyo

✉ chelak@iis.u-tokyo.ac.jp

📁 [ilia-chelak.github.io](https://github.com/ilia-chelak)

🌐 [ilia-chelak](#)

Experience

- Jan 2026 – **Research Intern**, *Microsoft Research Asia*, Tokyo, JP
Jun 2026 ◦ Work on a Vision-Language-Action (VLA) model without Vision-Language models (VLM).
- Oct 2023 – **Visiting Research Student**, *Osaka University Computer Vision Lab*, Osaka, JP
Feb 2024 ◦ Researched 3D reconstruction from multi-view images under Professor [Yasuyuki Matsushita](#).
◦ Introduced Structure-from-Motion initialization to neural SDF reconstruction, cutting training time from 30 min to **7 min** and capturing sub-millimeter details (see thesis).
- Mar 2022 – **Computer Vision Research Engineer**, *Watermarked*
Apr 2025 ◦ Achieved real-time resolution-independent watermarking for images and videos.
◦ Boosted message decoding accuracy for damaged images, videos, and audio by **60%**.
- Mar 2020 – **Volunteer Researcher**, *CVPR Lab*, *LUT University*, Lappeenranta, FI
Feb 2022 ◦ Collaborated on the [CoExist](#) ringed-seal re-identification project under Professor [Kälviäinen](#).
◦ Improved training speed by **10%** through an efficient loss function.
◦ Developed EDEN, a global pooling layer that raised ringed-seal image-retrieval accuracy by **2%** (see publications).

Education

- 2025 – **Ph.D. in Computer Vision**, *The University of Tokyo*, Tokyo, JP
present ◦ Focus: real-time multi-view 3D reconstruction • computer graphics
◦ Working at [Sato Lab](#) under the supervision of [Yoichi Sato](#) and [Yasuyuki Matsushita](#).
- 2022 – 2024 **M.Sc. in Data Science**, *University of Helsinki*, Helsinki, FI, GPA 4.8/5
◦ Focus: Computer Vision • Bayesian ML • Statistical inference
◦ [Thesis](#): “PriorSDF: Fast and Accurate 3D Shape Recovery by SfM-informed Neural SDFs”
- 2017 – 2021 **B.Sc. in Computer Science**, *St. Petersburg Polytechnic University*, St. Petersburg, RU, GPA 4.8/5
◦ Thesis: “Ringed seals re-identification with convolutional neural networks.”

Publications

Nepovinnikh, E., **Chelak, I.**, Immonen, V., Eerola, T., Kälviäinen, H., Kholiavchenko, M., Stewart, C. (2024). Species-Agnostic Patterned Animal Re-identification by Aggregating Deep Local Features. In *International Journal of Computer Vision*.

Nepovinnikh, E., **Chelak, I.**, Lushpanov, A., Eerola, T., Kälviäinen, H., & Chirkova, O. (2022). Matching individual Ladoga ringed seals across short-term image sequences. *Mammalian Biology*.

Chelak, I., Nepovinnikh, E., Eerola, T., Kälviäinen, H., & Belykh, I. (2021). EDEN: Deep Feature Distribution Pooling for Saimaa Ringed Seals Pattern Matching. In *Cyber-Physical Systems and Control II*, **Best Paper – 1st Runner-Up Award**.

Scholarships

- 2025 **Tokyo University SPRING GX scholarship**
◦ Stipend ¥180,000 / month + ¥360,000 / year research funds (~\$1,250 & ~\$2,500)

2023 **JASSO Student Exchange Support Program**

- Monthly stipend ¥80,000 (~\$550) per month for 6 months.

2022 **University of Helsinki scholarship**

- Full tuition fee waiver (worth €30,000; ~10% acceptance rate)

2020 **FIRST+ (Finnish–Russian) scholarship**

- FIRST+ mobility grant (€560 / month) for 5-month exchange

Skills

Computer Vision 3D reconstruction, image/video steganography, image retrieval, metric learning

Languages English (proficient), Russian (native), Japanese (intermediate)

References

Yasuyuki Matsushita

Senior Principal Research Manager,
Microsoft Research Asia Tokyo,
homepage: [click](#),
email: yasumat@ist.osaka-u.ac.jp,
relationship: Ph.D. co-supervisor

Yoichi Sato

Professor,
The University of Tokyo,
homepage: [click](#),
email: ysato@iis.u-tokyo.ac.jp,
relationship: Ph.D. co-supervisor